



Benedetta Mariani

Born: on 21 December 1995 | Nationality: Italian

📍 Department of Physics, University of Padua (Via Marzolo 8 Padua), Padova
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RESEARCH EXPERIENCES

October 2024 - present day

Post-doctoral researcher

Padova Neuroscience Center

Inference of directed interactions in the brain through fMRI data

Founded by the PNC Project INSIGHTS (PNRR Cascading Grant) within the Research and Innovation Program titled "MNESYS" (an integrated multiscale approach to the study of the nervous system in health and disease)

October 2020 - September 2024

PhD student in Neuroscience

Padova Neuroscience Center

Laboratory of Interdisciplinary Physics, Padova <https://www.liphlab.com/>

- Supervisor: Prof Samir Suweis
- Main research interests: combining the tools of physics of complex systems and data analysis to study collective behavior in Neuroscience and Systems biology (e.g. how information processing in neural systems is enhanced by phase transitions and synchronization phenomena)

December 2019 – September 2020

Ten months scholarship

Laboratory of interdisciplinary Physics, University of Padua

- Project title: "Criticality in rat somatosensory barrel cortex"
- Supervisors: Prof Samir Suweis and Prof. Stefano Vassanelli.

EDUCATION

December 2017 – October 2019

Master degree in Physics of Complex Systems

University of Turin, Italy

- Final grade: 110/110 cum laude
- Final project: "Signatures of criticality in rat brain: beyond avalanches statistics" performed at Liph Lab (Physics and Astronomy Dept, Padua) and at NeuroChip Lab (Dept. of Biomedical Sciences, Padua).

September 2015 – December 2017

Bachelor degree in Physics

University of Padua, Italy

- Final project: "Phase separation for intrinsically disordered proteins"

October 2014 – September 2015

Student of Biomedical Engineering

Polytechnic University of Milan

PUBLICATIONS

November 2023

Prenatal experience with language shapes the brain

- Science Advances Vol 9, Issue 47
- [Benedetta Mariani](#), Giorgio Nicoletti, Giacomo Barzon, Maria Clemencia Ortiz Barajas, Mohinish Shukla, Ramon Guevara, Samir Simon Suweis, Judit Gervain
- <https://www.science.org/doi/10.1126/sciadv.adj3524>; <https://www.aaas.org/news/babies-brains-are-primed-their-native-language-birth>

- July 2022 **Criticality and network structure drive emergent oscillations in a stochastic whole-brain model**
- Journal of Physics: Complexity, Volume 3, Number 2
 - Giacomo Barzon, Giorgio Nicoletti, [Benedetta Mariani](#), Marco Formentin, Samir Suweis
 - <https://iopscience.iop.org/article/10.1088/2632-072X/ac7a83/meta>
- June 2022 **Disentangling the critical signatures of neural activity**
- Scientific Reports volume 12, Article number: 10770
 - [Benedetta Mariani](#), Giorgio Nicoletti, Marta Bisio, Marta Maschietto, Stefano Vassanelli, Samir Suweis
 - <https://www.nature.com/articles/s41598-022-13686-0>
- August 2021 **Neuronal Avalanches Across the Rat Somatosensory Barrel Cortex and the Effect of Single Whisker Stimulation**
- Front. Syst. Neurosci., vol 15
- [Benedetta Mariani](#), Giorgio Nicoletti, Marta Bisio, Marta Maschietto, Roberto Oboe, Alessandro Leparulo, Samir Suweis, Stefano Vassanelli
- <https://www.frontiersin.org/articles/10.3389/fnsys.2021.709677/full>

TEACHING EXPERIENCE

- October 2025 **Teaching assistant**
- for the "Advanced Machine learning" module taught by prof. Nicolò Navarin at the Master in "Quantitative and Computational Biosciences" (UniPd)

THESIS CO-SUPERVISION

- December 2022 **Co-supervision of a UniPd Physics bachelor degree thesis**
- "Analyses of neural oscillations and synchrony in the brain"

TALKS, PRESENTATIONS AND POSTERS

- July 2025 **Poster Presentation at CNS 2025, Florence**
- Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches
- June 2025 **Contributed talk at the StatPhys satellite, Venice**
- February 2025 **Invited talk at the SIFS Young series seminar**
- Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches
- September 2024 **Poster presentation at Italian Network for computational Neuroscience, Rome**
- Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches
- May 2024 **Invited Talk, "Baby Rhythm" ERC closing workshop, Palazzo Bo, Padova, Italy**
- "Prenatal experience with language shapes the brain"
- February 2023 **Talk at Liph Lab Spring Workshop, Asiago (VI), Italy**
- "Collective oscillations in the rat barrel-thalamus network"
- November 2022 **Poster at Brain Criticality Hybrid Meeting November 7-9 2022, Online**
- "Collective oscillations in the rat barrel-thalamus network"
 - <https://braincriticality.org/2022/11/07/poster-2022-benedetta-mariani-collective-oscillations-in-the-rat-barrel-thalamus-network/>

September 2022	Poster at Bernstein Conference 2022, Berlin • "Collective oscillations in the rat barrel-thalamus network" • https://abstracts.g-node.org/conference/BC22/abstracts#/uuid/83d6de84-d449-4f47-ba63-7cd486e93e40
31st March 2022	Talk at Liph Lab Spring Workshop, Asiago (VI), Italy • "Modelling oscillations in the rat barrel-thalamus network"
September 2021	Poster at Bernstein Conference 2021, online • "Disentangling the role of external and intrinsic dynamics on the critical signatures of neural activity" • https://abstracts.g-node.org/conference/BC21/abstracts#/uuid/00a75410-d18a-4ba6-a14d-aa682e2c8777
31st May-12nd June 2021	Poster at the Beg Rohu Summer School, St. Pierre Quiberon, France • "On the critical signatures of neural activity"
6th October-9th October 2020	Contributed talk at Brain Criticality Virtual Meeting, online • "Brain criticality beyond resting state neuronal avalanches" • https://braincriticality.org/2020/10/11/benedetta-mariani-brain-criticality-beyond-resting-state-neuronal-avalanches-spotlight-talk/
29th September - 1st October 2020	Poster at Bernstein Conference 2020, online • "Brain criticality beyond resting state neuronal avalanches" • https://abstracts.g-node.org/abstracts/b9ec73de-f9db-4df2-9ab1-c08c2c532e65
16th - 18th January 2020	Talk at Liph Lab winter Workshop, Folgaria (TN) • I presented my research outputs.

ATTENDED CONFERENCES AND SCHOOLS

28-31 May 2024	Stochastic Models and Experiments in Ecology and Biology, Gran Sasso Science Institute, L'Aquila, Italy
May 2024	"Baby Rhythm", Judit Gervain ERC closing workshop, Palazzo Bo, Padova
November 2022	Brain Criticality Hybrid Meeting November 7-9 2022, Online
September 2022	Bernstein Conference 2022, Berlin
16th-20th March 2022	Cosyne Tutorial and Cosyne Main meeting, Lisbon
15th-24th September 2021	EITN Fall School in Computational Neuroscience, Paris • I took part in the project group "Mean field and population models" and presented the outputs of our project. • I attended lectures and tutorials on different models and softwares to simulate networks of neurons at different levels of complexity (Brian, Nest, Neuron, TVB)
September 2021	Bernstein Conference 2021, online
31st May - 12nd June 2021	The Beg Rohu Summer School, St. Pierre Quiberon, France
23rd February - 26th February 2021	Computational and Systems Neuroscience (Cosyne) 2021, online
14th - 16th December 2020	Brain Hack Padova, online • I led the project group "Association of Microstructural white matter and personality traits based on Human connectome project dataset"

6th October - 9th October 2020	Brain Criticality Virtual Meeting, online
29th September - 1st October 2020	Bernstein Conference 2020, online
18th - 23th July 2020	29th Annual Computational Neuroscience Meeting, Organization for Computational Neuroscience, online
29th June - 3rd July 2020	Youth in High-dimensions: Machine Learning, High-dimensional Statistics and Inference for the New Generation, online
9th - 14th September 2019	Summer School in Computational and Theoretical Models in Neuroscience, European centre for living technology, Venice

AWARDS

November 2023	We published "Prenatal experience with language shapes the brain" https://www.science.org/doi/10.1126/sciadv.adj3524?utm_campaign=SciMag&utm_source=Twitter&utm_medium=ownedSocial in the high impact journal "Science Advances" https://www.aaas.org/news/babies-brains-are-primed-their-native-language-birth and gained visibility (journalists from many countries quoted and interviewed us, including "Le Scienze" journal, "Nature Italy" journal, "El País, Spain", "The Times, UK", "Le Figaro, France", "New Scientist, UK", e.g.: https://www.lescienze.it/news/2023/11/23/news/linguaggio_mamma_cervello_neonati-14259373/ ; https://www.nature.com/articles/d43978-023-00181-x ; and we participated to podcasts broadcasting our work, e.g: https://www.thenakedscientists.com/articles/interviews/babies-learn-language-they-are-even-born
2022	Our paper "Disentangling the critical signatures of neural activity" https://www.nature.com/articles/s41598-022-13686-0 has been included in the Top 100 Neuroscience paper of 2022 Scientific Reports Journal.
2021	Winner of SECS (Scholarships for Events on Complex Systems) grant, provided by the Young Researchers of the Complex Systems Society.
2014	Winner of the award for the high school students with the highest final grade and honors, granted by MIUR, the Italian Ministry for Education, University and Research.

COMPUTER SKILLS

Programming Languages	• Python: advanced level • C++, Mathematica, R, matlab: intermediate level
Operating systems	• Windows: advanced level • Unix: advanced level

PERSONAL SKILLS

Languages known	• Mother tongue: Italian • English: level B2 (First Certificate of English) • German: level A2 (certificate from the Goethe Institute of Bremen)
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